Best Practices for Running

Microsoft SQL Server on AWS

Updated June 2026 · Covers SQL Server 2022 – 2025 · Amazon RDS · EC2 · RDS Custom

256 TiB	448 vCPU	24 TB	256K IOPS
RDS Max Storage	EC2 Max vCPUs	EC2 Max RAM	RDS Custom IOPS
4× increase (Dec 2025)	U-series / bare metal	U7in-32tb series	io2 Block Express (2025)

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Fully managed database service — AWS handles patching, backups, HA, and scaling.

- Up to **128 vCPU**
- Up to **256 TiB storage** (Dec 2025 expansion)
- Up to **4 TiB RAM** (db.x2idn)
- License-Included or RDS Custom BYOL

SQL Server on Amazon EC2

Self-managed virtual machine — full OS and SQL Server control, BYOL or License-Included AMIs.

- Up to **448 vCPU** (bare-metal U-series)
- Up to **24 TB RAM** (U7in-32tb)
- Up to **400 TB storage** with EBS
- SQL Server 2025 AMI available Nov 2025

EC2 vs. RDS — Feature Comparison (2026)

Feature	EC2	RDS
License Included AMI	✓	✓
BYOL	✓	RDS Custom only
Full OS/instance control	✓	✗
Automated patching	Manual	✓
Multi-AZ (AWS-managed)	Self-managed	✓
Read scale-out replicas	Self-managed	✓ (Ent. 2016+)
Optimize CPU (SMT control)	✓	✓ M7i/R7i (Dec 2025)
SQL Server 2025 support	✓ (BYOL + LI AMI)	Roadmap
Storage up to 256 TiB	✓ (EBS)	✓ (Dec 2025)

02 Amazon RDS for SQL Server

Amazon RDS removes the undifferentiated heavy lifting of database administration. AWS manages hardware, OS patching, DBMS patching, automated backups, and Multi-AZ high availability — leaving teams free to focus on schema design, query tuning, and application logic.

Supported Versions & Editions (2026)

Attribute	Details
SQL Server Versions	2016 · 2017 · 2019 · 2022
Editions	Express · Standard · Enterprise (Web edition retired in SQL Server 2025)
High Availability	AlwaysOn AG (2016+) · DB Mirroring (2016 legacy)
Encryption	TDE · Column-level · Always Encrypted · TLS · EBS
Authentication	Windows Auth · SQL Auth · IAM DB Auth (via AD)
Backups	AWS-managed automated + Native .BAK to/from S3
BI Stack	SSRS, SSIS (see RDS limitations guide for SSAS)
Optimize CPU	M7i and R7i families (launched Dec 2025)

AWS-Managed vs. Customer-Managed Responsibilities

AWS Manages

- Power, HVAC, Networking
- OS Install & Patching
- DBMS Install & Patching
- Database Backups
- Multi-AZ Failover
- Storage Auto-scaling

Customer Manages

- Schema & index design
- Query & workload tuning
- User & permission management
- Application connection strings
- Parameter group configuration
- Point-in-time restore strategy

03 RDS — Compute, Storage & Scaling

Recommended Instance Families for RDS SQL Server (2025-2026)

Family	Use Case	Key Spec	Notes
db.t3 / t3g	Dev/Test, low traffic	2–8 vCPU, burst	Cost-optimal for non-prod
db.m7i	General purpose	2–96 vCPU	Balanced compute & memory
db.r7i	Memory-intensive	2–96 vCPU, high RAM	Optimize CPU eligible
db.m7i + Opt. CPU	License savings	Custom vCPU count	Up to 55% lower cost
db.x2idn	Largest SQL workloads	128 vCPU, 4 TiB	Max RDS memory

NEW — Optimize CPU for RDS SQL Server (Dec 2025)

Amazon RDS for SQL Server on M7i and R7i instances now supports **Optimize CPU**, which disables Simultaneous Multi-Threading (SMT) to halve the active vCPU count while preserving full physical core performance. Because RDS SQL Server pricing is license-inclusive and core-based, disabling SMT can reduce costs by **up to 55%** compared to equivalent 6th-gen instances.

Storage Options

Volume Type	Max Size	Max IOPS	Max Throughput	Best For
gp3 (General Purpose SSD)	256 TiB*	16,000	1,000 MiB/s	Standard workloads
io2 (Provisioned IOPS SSD)	256 TiB*	64,000	4,000 MiB/s	Mission-critical OLTP
io2 Block Express (Custom)	64 TiB	256,000	4,000 MiB/s	Largest OLTP/DW

** Up to 256 TiB via additional storage volumes (up to 3 extra volumes) — announced Dec 2025.*

Backup: Native .BAK to Amazon S3

RDS SQL Server supports native backup/restore directly to S3 using the **sqlserver-backup-restore** option group. Compression is supported. Only full and differential backups are available natively; transaction log backups are AWS-managed. Multi-file backup/restore is supported for parallel throughput.

04 High Availability on Amazon RDS

Multi-AZ Deployment — AlwaysOn AG

RDS Multi-AZ for SQL Server 2016, 2017, 2019, and 2022 uses **AlwaysOn Availability Groups** (Basic AG for Standard Edition). The secondary is a hot synchronous standby with automatic failover. Read traffic is not routed to the secondary in standard Multi-AZ.

HA Mode	Versions	Replication	Failover
AlwaysOn AG	SQL 2016 / 2017 / 2019 / 2022	Synchronous	Automatic
Basic AG (Std Ed)	SQL 2016+ Standard	Synchronous	Automatic
DB Mirroring	SQL 2016 (legacy)	Synchronous	Automatic

Read Scale-Out (Enterprise Edition 2016+)

Up to **5 asynchronous read replicas** in-region for Enterprise Edition. Each replica has a separate endpoint. Replicas can be promoted to standalone instances. Cross-region read replicas are not supported on RDS SQL Server.

RDS Custom for SQL Server — Extended Control

RDS Custom (BYOL) — When to Use

- Full OS and SQL Server access required (e.g. custom CLR, linked servers, xp_cmdshell)
- io2 Block Express: up to 64 TiB and 256,000 IOPS (GA Jan 2025)
- Bring-your-own SQL Server license (SA required)
- Agent customization, Windows-level tuning, custom monitoring agents

05 Monitoring & Performance Insights

1. **Amazon CloudWatch Metrics**

   CPU, connections, free storage, IOPS, latency, throughput, swap — collected at hypervisor level. Set alarms and dashboards.

2. **Enhanced Monitoring**

   26 OS-level and per-process metrics delivered to CloudWatch Logs at up to 1-second granularity. Agent-based — closer to true CPU than hypervisor.

3. **Performance Insights**

   Visualizes DB load by wait type, SQL, user, host. Default 7-day retention (paid: 2 years). Identify top SQL and waits at a glance.

4. **SQL Server Native (DMVs/DMFs)**

   sys.dm_exec_query_stats, sys.dm_os_wait_stats, Query Store, Extended Events, sys.dm_db_index_usage_stats — full T-SQL visibility.

5. **Third-Party & Open Source**

   Ola Hallengren maintenance scripts, SolarWinds DPA, Redgate SQL Monitor, Datadog, SentryOne (SQL Sentry) — integrate via CloudWatch or agent.

Key CloudWatch Metrics — Quick Reference

`CPUUtilization`	`FreeableMemory`	`ReadIOPS / WriteIOPS`
`DatabaseConnections`	`FreeStorageSpace`	`ReadLatency / WriteLatency`
`DiskQueueDepth`	`SwapUsage`	`NetworkReceive/TransmitThroughput`

06 SQL Server on Amazon EC2

For workloads requiring full OS control, Windows clustering, custom configurations, or SQL Server 2025, Amazon EC2 is the deployment target. Choose from License-Included AMIs or bring your own SQL Server license (BYOL).

EC2 Instance Families for SQL Server (2025-2026)

Family	Generation	Best For SQL Server	Highlight
M8i / M8i-flex	8th Gen Intel Xeon 6	General purpose SQL workloads	15% better price-perf vs M7i; 2.5× memory BW
R8i / R8i-flex	8th Gen Intel Xeon 6	Memory-intensive OLTP, large buffer pools	32 vCPU / 256 GB matches SQL 2025 Standard limit
M7i / R7i	7th Gen Intel Sapphire	Balanced / memory workloads	RDS Optimize CPU eligible
C8i	8th Gen Intel Xeon 6	CPU-bound batch workloads	High clock speed, AVX-512
I8g / I8ge	8th Gen Graviton4	NVMe tempdb hosting	High-IOPS local NVMe SSD
X8i / U7i-*	Ultra-memory	Very large in-memory DW	Up to 24 TB RAM
T3 / T4g	Burstable	Dev/Test, low-traffic	Credit-based CPU bursting

Getting Started on EC2

Option A — AWS-Provided AMI (License Included)

- Launch from AWS Marketplace. SQL Server 2019, 2022, and 2025 AMIs available on Windows Server 2022. License cost rolled into EC2 hourly rate.

Option B — BYOL (Bring Your Own License)

- Spin up a Windows EC2 instance, install SQL Server from your media. Requires Software Assurance for cloud deployment rights. AWS License Manager helps track core-based license compliance.

Option C — AWS Launch Wizard

- Point-and-click SQL Server deployment (standalone or AlwaysOn AG). Supports SQL Server Developer Edition in 2025 without MSDN.

07 Optimizing EC2 Performance for SQL Server

EBS Volume Selection

Volume	Durability	Max IOPS/vol	Max Throughput/vol	Use Case
gp3	99.8–99.9%	16,000	1,000 MiB/s	Data/log files, balanced cost
io2	99.999%	64,000	4,000 MiB/s	Mission-critical OLTP
io2 Block Express	99.999%	256,000	4,000 MiB/s	Largest DW/OLTP
st1 (HDD)	99.8%	500	500 MiB/s	Backup files only

Disk Layout Best Practices

File Type	Access Pattern	Recommended Volume	Allocation Unit
Data Files (.mdf / .ndf)	Random	gp3 or io2	64 KB
Transaction Log (.ldf)	Sequential	gp3 or io2	64 KB
TempDB Data	Random (heavy)	NVMe Instance Store	64 KB
TempDB Log	Sequential	NVMe Instance Store	64 KB
Backups (.bak)	Sequential write	st1 / S3 native backup	64 KB

TempDB on NVMe Instance Store

Instance types such as **M8idb, R8idb, I8g/I8ge, z1d** include direct-attached NVMe SSD storage — included in the hourly price — delivering extremely low latency I/O without consuming EBS bandwidth. Mirror instance store volumes for tempdb resilience across a failover.

Amazon FSx for Windows File Server

FSx for Windows provides a fully managed SMB file system for SQL Server data files via UNC paths. Ideal for Failover Cluster Instances (FCI) requiring shared storage without managing SAN. Create databases referencing the FSx UNC path directly in T-SQL.

```
CREATE DATABASE [MyDB] ON PRIMARY (FILENAME = N'\\fsx-dns\share\MyDB.mdf'…)
```

Optimize CPU — EC2 (License Cost Reduction)

On EC2, **Optimize CPU** lets you disable hyperthreading (SMT) to reduce the vCPU count seen by SQL Server, lowering core-based Windows and SQL Server licensing costs while maintaining the same physical core count and near-equivalent performance. Particularly effective for memory-heavy readable secondaries on R8i instances.

08 HA/DR Best Practices for SQL Server on EC2

HA/DR Options Summary

Solution	Scope	Failover	Key Benefit
AlwaysOn AG (Multi-AZ)	Database-level	Auto/Manual	Granular per-DB HA; readable secondaries
AlwaysOn AG (Multi-Region)	Database-level	Manual	Cross-region DR via async replica + Transit GW
Contained AG (SQL 2025)	Database+instance-level	Auto/Manual	Replicates logins, SQL Agent jobs automatically
SQL FCI + Amazon FSx	Instance-level	Auto (WSFC)	Instance-level HA; logins/jobs survive failover
Log Shipping	Database-level	Manual	Simple, low-cost warm DR
Distributed AG (DAG)	Cross-AG	Manual	Migration & multi-site; improved in SQL 2025

AlwaysOn AG — Best Practices (Multi-AZ)

- Deploy replicas across at least 2 Availability Zones
- Use a File Share Witness (FSW) in a 3rd AZ or separate subnet
- Set **MultiSubnetFailover=Yes** in connection strings
- Always connect via the AG Listener — never direct to replica endpoint
- Set **RegisterAllProvidersIP=Yes** and reduce **HostRecordTTL**
- Use SQL Server Native Client or latest ODBC/OLE DB provider
- Enable Readable Secondaries for reporting to offload primary
- SQL Server 2025: offload full/diff/t-log backups to secondary replicas

Contained Availability Groups — SQL Server 2025 New Feature

Contained AG simplifies instance-level replication

- Automatically replicates logins, SQL Agent jobs, and certificates between nodes
- Replaces need for manual scripting of server-level objects to secondaries
- Contained DAG: master and msdb system databases included in the AG
- Simplifies migrations from on-premises to EC2 and between AWS regions

09 SQL Server 2025 — New Features on AWS

Microsoft released SQL Server 2025 for general availability on **November 18, 2025**. AWS-provided License-Included AMIs and BYOL support are available immediately on EC2. RDS support is on the roadmap.

Edition Changes

Standard Edition limits raised to **32 CPU cores and 256 GB RAM**. Web Edition removed. Standard and Enterprise Developer Editions added for non-prod use without MSDN.

Resource Governor in Standard Edition

Resource Governor (previously Enterprise-only) now available in Standard. Limits CPU, memory, IOPS, and concurrent requests per workload group. Also governs tempdb space usage per workload group.

Secondary Replica Backups

Full, differential, and transaction log backups can now run from AlwaysOn AG secondary replicas, offloading storage and CPU from the primary. Combine with S3 native backup for complete cloud-native backup strategy.

ZSTD Backup Compression

New ZSTD algorithm delivers higher compression ratios than the legacy MS_XPRESS algorithm, reducing S3 backup storage footprint and transfer costs.

Contained Availability Groups

Server-level objects (logins, SQL Agent jobs, linked servers, certificates) automatically replicate between AG nodes. Replaces Distributed AGs for most migration and multi-site scenarios.

Query Store for Readable Secondaries

Enabled by default. Query stats from readable secondaries flow back to the primary and are persisted, enabling consistent execution plans across all replicas — critical for scaled-out read workloads.

ADR in TempDB

Accelerated Database Recovery in tempdb provides instant transaction rollback and aggressive log truncation — prevents tempdb log space exhaustion under long-running or runaway transactions.

REST / AI Integration

sp_invoke_external_rest_endpoint enables T-SQL calls to AWS services (Amazon Bedrock, S3, Lambda) directly, without external code. Enables in-database AI workflows and event-driven patterns.

10 Migration Pathways to AWS

Migrating to Amazon RDS

1. Native .BAK Backup/Restore

- Most common approach. Take a full backup, upload to S3, restore via sqlserver-backup-restore option group. Supports compression and multi-file.

2. AWS Database Migration Service (DMS)

- Minimize downtime with continuous CDC replication. Supports homogeneous (SQL→SQL) and heterogeneous migrations. Schema Conversion Tool (SCT) assists cross-platform moves.

3. SQL Server Replication

- Push transactional replication subscription to RDS target. Zero-downtime cutover; publisher remains on-premises.

4. AWS Marketplace Tools

- Third-party ETL, sync, and migration tools available. ApexSQL, Striim, Attunity (now AWS DMS Studio), etc.

Migrating to / Between EC2

1. Backup & Restore

- SQL Server native backup to S3 (SQL 2022+ supports S3 natively), or via Windows file copy. Simple and reliable for large databases.

2. AlwaysOn AG (Distributed AG for SQL 2016+)

- Seed an async replica on EC2, then cut over. Distributed AGs enable cross-network seeding. Improved in SQL Server 2025 with reduced network saturation in async mode.

3. AWS Application Migration Service (MGN)

- Lift-and-shift entire Windows SQL Server instance to EC2. Block-level replication with near-zero RPO cutover.

Migration Decision Matrix

Scenario	Recommended Tool	Downtime
On-prem SQL → RDS (same version)	Backup/Restore (.BAK to S3)	Hours (backup window)
On-prem SQL → RDS (cross-version)	DMS + SCT	Minutes (CDC)
On-prem SQL → EC2 (lift & shift)	AWS MGN	Minutes (cutover)
On-prem SQL → EC2 (live replication)	AlwaysOn AG / Distributed AG	Seconds
EC2 → RDS (schema compatible)	Backup/Restore or DMS	Minutes–Hours
SQL Server 2019 → SQL Server 2025 on EC2	In-place upgrade or parallel AG	Planned window

For migration support, visit: **aws.amazon.com/sql-server** or contact your AWS account team.